Feasibility Study
Real-Time Sign Language Detection and Translation System Using AI

1. Introduction

A feasibility study evaluates whether a proposed project can be successfully developed and implemented. It considers the economic, technical, operational, and schedule aspects before development begins. The proposed Real-Time Sign Language Detection and Translation System uses Artificial Intelligence, Computer Vision, and Deep Learning to recognize sign language gestures through a webcam and convert them into text and speech. The system aims to improve communication between hearing-impaired individuals and the general public while providing an affordable and accessible solution.

2. Economic Feasibility

The project is economically feasible because it uses open-source technologies such as Python, OpenCV, MediaPipe, TensorFlow/PyTorch, and Streamlit, eliminating software licensing costs. Only a standard computer and webcam are required. The system has low development and maintenance costs, making it suitable for educational institutions and organizations.

Item	Estimated Cost (INR)
HD Webcam (Optional)	■2,000–3,000
Cloud GPU (Optional)	■2,000–5,000
Internet & Storage	■500–1,000
Miscellaneous	■500
Total	■5,000–9,500

3. Technical Feasibility

The project uses well-established AI and computer vision technologies. MediaPipe detects hand landmarks, OpenCV processes live video, and TensorFlow/PyTorch performs gesture recognition. The recognized gestures are translated into text and speech using a Text-to-Speech engine. The required software is freely available and supported by large developer communities. Standard hardware with an Intel i5 processor, 8 GB RAM, and an HD webcam is sufficient for development and testing.

4. Operational Feasibility

The application is simple to operate. Users perform sign language gestures in front of a webcam, and the system instantly converts them into text and speech. The application can be used in schools, hospitals, banks, government offices, customer service centers, and public places. It improves accessibility and reduces communication barriers while requiring minimal user training.

5. Schedule Feasibility

The project can be completed within 15 weeks. The work includes requirement analysis, system design, dataset preparation, model development, system integration, testing, and documentation. A structured schedule ensures that each phase is completed on time while providing sufficient time for debugging and evaluation.

6. Conclusion

The proposed Real-Time Sign Language Detection and Translation System is feasible from economic, technical, operational, and scheduling perspectives. It combines affordable hardware, open-source software, and modern AI techniques to provide an efficient communication solution for hearing-impaired individuals. The project is practical, cost-effective, scalable, and suitable as a B.Tech final-year project.

Schedule Feasibility

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Project Schedule

Phase	Activity	Duration	Weeks
Phase 1	Requirement Analysis & Literature Review	2 Weeks	Week 1–2
Phase 2	System Design & Architecture	2 Weeks	Week 3–4
Phase 3	Dataset Preparation & Preprocessing	2 Weeks	Week 5–6
Phase 4	Model Development & Implementation	4 Weeks	Week 7–10
Phase 5	System Integration	2 Weeks	Week 11–12
Phase 6	Testing, Debugging & Evaluation	2 Weeks	Week 13–14
Phase 7	Documentation & Final Review	1 Week	Week 15

15-Week Gantt Chart

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Requirement Analysis	■	■													
System Design			■	■											
Dataset Preparation					■	■									
Model Development							■	■	■	■					
System Integration											■	■			
Testing & Debugging													■	■	
Documentation & Final Review															■

Legend: ■ = Work scheduled during that week.